1. Notation: let e_{ij} denote the elementary matrix with a 1 in the i, j-th position and zeroes elsewhere
(a) Show that the exponential map for $SL(2, \mathbb{C})$ is not surjective.

Proof. Let $A \in \exp(\mathfrak{sl}(2, \mathbb{C}))$ so that $A = \exp(B)$, then since we are working in $\mathbb{C}$ we can conjugate B to its Jordan canonical form $J = PBP^{-1}$, the jordan canonical form of a matrix with trace zero is one of

$$J_1 = \begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix} \qquad J_2 = \begin{pmatrix} \lambda & 1 \\ 0 & -\lambda \end{pmatrix}$$

The exponential of either Jordan canonical form looks like:

$$\begin{aligned} \exp(J_1) &= \begin{pmatrix} e^{\lambda} & 0 \\ 0 & e^{-\lambda} \end{pmatrix} \\ \exp(J_2) &= \exp(J_1 + e_{12}) = \exp(J_1)\exp(e_{12}) = \begin{pmatrix} e^{\lambda} & 0 \\ 0 & e^{-\lambda} \end{pmatrix}(1 + e_{12}) = \begin{pmatrix} e^{\lambda} & e^{\lambda} \\ 0 & e^{-\lambda} \end{pmatrix} \end{aligned}$$

So that $\exp(J_2)$ has canonical form $\begin{pmatrix} 1 & 1 \\ 0 & e^{-2\lambda} \end{pmatrix}$. Now since $A = \exp(B) = P\exp(J)P^{-1}$, A must have one of these two normal forms. It follows that

$$\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$$

is not in the image of exp. □

(b) Show the exponential map for $SL(2, \mathbb{R})$ is not surjective

Proof. Assume it were, then $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$ would be in the image of the exponential map on $\mathfrak{sl}(2, \mathbb{R})$, since the exponential map on $\mathfrak{sl}(2, \mathbb{R})$ is the restriction of the exponential map on $\mathfrak{sl}(2, \mathbb{C})$ to real matrices, this contradicts (a). □

2. **(a)** Show that $\mathrm{Mat}_n(\mathbb{C}) \hookrightarrow \mathrm{Mat}_{2n}(\mathbb{R})$ is a subalgebra.

Proof. by providing a map $\phi : \mathbb{C} \hookrightarrow \mathrm{Mat}_2(\mathbb{R})$ we can extend this to $\mathrm{Mat}_n(\mathbb{C})$ by taking a complex matrix to its block form pointwise. Explicitly the map on matrices is $\psi : (z_{nm})_{n,m} \mapsto (\phi(z_{nm}))_{n,m}$.

Suppose $z \in \mathbb{C}$, we can define $\psi(z)$ by taking z to its multiplication action on $\mathbb{R} \oplus i\mathbb{R}$, concretely

$$\phi : a + bi \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

Injectivity, $\mathbb{R}$-linearity and $1 \mapsto 1$ are immediate. To see that multiplication is preserved note that the matrix in the image is the action of multiplication on $\mathbb{R} \oplus i\mathbb{R}$, so composition is equivalent to multiplication by the product. □

(b) Show that $U(n) = GL(n, \mathbb{C}) \cap O(2n)$

Proof. It suffices to show that the inclusion of algebras above maps $U(n)$ into $O(2n)$. We first note that $\phi(\overline{z}) = \phi(z)^{\mathrm{T}}$ for $z \in \mathbb{C}$. Now let $A \in U(n)$, then we work in block form and use the identity just stated relating transposition and conjugation to get

$$\psi(A^{\dagger}) = (\phi(z_{mn})^{\mathrm{T}})_{n,m} = \psi(A)^{\mathrm{T}}$$

Now we apply the fact that ψ is an algebra morphism to conclude that

$$\psi(A)\psi(A)^T = \psi(A)\psi(A^{\dagger}) = \psi(AA^{\dagger}) = \psi(1) = 1$$

□

3. (a) Show the following map is an algebra morphism

$$\begin{aligned} \mathbb{H} &\to \mathrm{Mat}_2(\mathbb{C}) \\ a + ib + jc + kd &\mapsto \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix} \end{aligned}$$

Proof. $1 \mapsto 1$ is immediate, moreover $\mathbb{R}$-linearity of the matrix entries ensures $\mathbb{R}$-linearity of the morphism, we are only left to check the multiplicative structure is preserved. Similar to the methodology of the previous problem, we can provide an interpretation of this map to avoid matrix pushing. To see this note that we can write $\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$, then the map above takes an element $x \in \mathbb{H}$ to its action by right multiplication on this vectorspace decomposition. Thus the multiplication being preserved simply follows from associativity since the composition of multiplication actions is the action of the product. $\square$

(b) Show the image of the unit quaternions under this map is $SU(2)$

Proof. Lets call the algebra morphism ϕ, we notice that $\phi(\overline{x}) = \phi(x)^\dagger$, so that if $x\overline{x} = 1$, then

$$\phi(x)\phi(x)^\dagger = \phi(x\overline{x}) = \phi(1) = 1$$

So that unit quaternions map into $U(2)$, moreover $\det(\phi(x)) = ||x||$, so that unit quaternions map to matrices of determinant one. We need only check the map is surjective, we use that $SU(2)$, and S^3 are both 3 dimensional manifolds, and that the map ϕ is an embedding $S^3 \hookrightarrow SU(2)$, hence also a local diffeomorphism. Since S^3 is compact, $\phi(S^3)$ is compact and hence closed so $SU(2)$ being connected allows us to conclude that ϕ is surjective. $\square$

(c) Prove that $Sp(1) \cong SU(2)$ as Lie groups.

Proof. $Sp(1) \subset \mathbb{H}$ are the quaternions such that $||ax|| = ||x||$ for all other quaternions x. These are exactly the unit quaternions. ϕ being a morphism of algebras from part (a) implies in particular it is a group homomorphism, and a diffeomorphism of manifolds from part (b) implies it is a Lie group isomorphism. $\square$

4. Give an explicit covering of $SO(4)$ by $SU(2) \times SU(2)$.

Proof. $SO(4)$ are the linear maps which map $S^3 \to S^3$, moreover by linearity each element of $SO(4)$ is determined by its values on S^3. Using this identification we may define maps in $SO(4)$ as maps on the unit quaternions, namely define

$$\begin{aligned} F : SU(2) \times SU(2) &\to SO(4) \\ (A, B) &\mapsto (X \mapsto AXB^\dagger) \end{aligned}$$

where we are identifying unit quaternions to matrices in $SU(2)$ as in the previous question.

This map is smooth, and its image indeed lies in $O(4)$, to see this, the action preserves norm since A, B are unitary,

$$AXB^\dagger(AXB^\dagger)^\dagger = AXB^\dagger BX^\dagger A^\dagger = AX1X^\dagger A^\dagger = A1A^\dagger = 1$$

And to see that the image lies in the connected component $\det(X) = 1$, we use that the idenitity map is the image of $(1, 1)$, and that $SU(2) \times SU(2)$ is connected, so its image lies in the connected component of the identity $SO(4)$. To see that this defines a covering map, we first check that F is a morphism of lie groups, i.e. that $F(A, B)F(C, D) = F(AC, BD)$, this is immediate since for any $X \in S^3 \subset \mathbb{H} \subset \mathrm{Mat}_2(\mathbb{C})$ we have

$$A(CXD^\dagger)B^\dagger = ACX(BD)^\dagger$$

Whence we will be done by invoking (problem 7) so long as we can show $d_{(1,1)}F : \mathfrak{su}(2) \oplus \mathfrak{su}(2) \to \mathfrak{so}(4)$ is bijective. To see this, we consider $(\exp(\xi), \exp(\eta)^\dagger)$ acting on $X \in S^3$, but since $\exp(t\eta) \in SU(2)$, we have $\exp(t\eta)^\dagger = \exp(-t\eta)$, so that we may compute

$$\begin{aligned} d_{(1,1)}F &: \mathfrak{su}(2) \oplus \mathfrak{su}(2) \to \mathfrak{so}(4) \\ &(\xi, \eta) \mapsto \left(X \mapsto \frac{d}{dt}\bigg|_{t=0} \exp(t\xi) X \exp(-t\eta) \right) \end{aligned}$$

We may compute this using the definition of the exponential map on matrix lie algebras $\exp(tA) = 1 + tA + \mathcal{O}(t^2)$, pluginh in this expansion, we get

$$d_{(1,1)}F(\xi, \eta) = \left(X \mapsto \frac{d}{dt}\bigg|_{t=0} \left(X + t\xi X - tX\eta + \mathcal{O}(t^2) \right) \right) = (X \mapsto \xi X - X\eta)$$

By plugging in $X = 1$, we see that for (ξ, η) to be in the kernel, we require that $\eta = \xi$, and thus $\ker d_{(1,1)}F$ is a subset of the center of $\mathfrak{su}(2)$, which is trivial. Thus $d_{(1,1)}F(\xi, \eta)$ is injective, moreover $\dim \mathfrak{su}(2) = 3$, and $\dim \mathfrak{so}(4) = 6$, so that by counting dimensions $d_{(1,1)}F$ is bijective. This suffices to show by (problem 7) that F is a covering map, to see that the index is 2, we compute the kernel. Suppose that $AXB^\dagger = X$ for all $X \in S^3 = SU(2)$, then plugging in $X = 1$, we find $AB^\dagger = 1$, so that $A = B$, it follows that $AXA^\dagger = XAA^\dagger = X$ for all $X \in SU(2)$, so that A is in the center of $SU(2)$, hence $A = \pm 1$, and since both of the resulting elements are in the kernel from this we conclude that $\ker \phi = \{(1, 1), (-1, -1)\}$ and that the cover has degree 2. $\square$

5. Show that for matrices A, B,

$$\log(\exp(tA)\exp(tB)\exp(-tA)\exp(-tB)) = t^2[A, B] + \mathcal{O}(t^3)$$

Proof. log and exp are locally inverse, so we can rewrite the above as:

$$\log(\exp(tA)\exp(tB)\exp(-tA)\exp(-tB)) = \log(e^{\log(\exp(tA)\exp(tB))} e^{\log(\exp(-tA)\exp(-tB))})$$

Then, the Baker-Campbell-Hausdorff formula tells us that the terms $\eta := \log(\exp(tA)\exp(tB)) = tA + tB + \frac{t^2}{2}[A, B] + \mathcal{O}(t^3)$, and similarly $\xi := \log(\exp(-tA)\exp(-tB)) = -tA - tB + \frac{t^2}{2}[A, B] + \mathcal{O}(t^3)$, then we can reapply the BCH formula to these expansions to get that

$$\begin{aligned} \log(\exp(tA)\exp(tB)\exp(-tA)\exp(-tB)) &= \log(e^{\log(\exp(tA)\exp(tB))} e^{\log(\exp(-tA)\exp(-tB))}) \\ &= \log(\exp(\eta)\exp(\xi)) \\ &\overset{\text{BCH}}{=} (tA + tB + \frac{t^2}{2}[A, B] + \mathcal{O}(t^3)) + (-tA - tB + \frac{t^2}{2}[A, B] + \mathcal{O}(t^3)) \\ &+ \frac{1}{2}[\eta, \xi] + \mathcal{O}(t^3) \\ &= t^2[A, B] + \frac{1}{2}[\eta, \xi] + \mathcal{O}(t^3) \end{aligned}$$

So it only remains to check that $[\eta, \xi] \in \mathcal{O}(t^3)$. We check this explicitly:

$$\begin{aligned} [\eta, \xi] &= (tA + tB)(-tA - tB) + \mathcal{O}(t^3) - (-tA - tB)(tA + tB) + \mathcal{O}(t^3) \\ &= -(tA + tB)^2 + (tA + tB)^2 + \mathcal{O}(t^3) \\ &= \mathcal{O}(t^3) \end{aligned}$$

$\square$

6. Let G be a connected Lie group, and U an open neighborhood of the group unit e. Show that any $g \in G$ can be written as a product of elements $g_1 \cdot g_2 \cdots g_n$ with $g_j \in U$.

Proof. Denote inv as the inverse map, and L_g as the map by left multiplication by a group element $g \in G$. Since inv is a diffeomorphism $\text{inv}(U)$ is an open subset of G, moreover $e \in U \cap \text{inv}(U)$, so it will suffice to show that all elements of G can be written as products of elements in $V := U \cap \text{inv}(U)$.

Now, define $S = \{\prod_1^n g_j \mid n \in \mathbb{Z}_{>0} \text{ and } g_j \in V\}$, to see that S is a subgroup of G, we note that $e \in S$, S is closed under products by definition, and if $g \in S$, then $g = \prod_1^n g_j$ with g_j in V, so that $\prod_1^n g_{n+1-j}^{-1} \in S$ since $V = \text{inv}(V)$. Moreover, $S \subset G$ is open, this can easily be seen since $S = \bigcup_{g \in S} L_g(V)$, where each $L_g(V)$ is open since L_g is a diffeomorphism. Now, since S is open to see that $S = G$ it will suffice to prove that S is also closed since G is connected. To see that S is closed, we prove that S^c is open. This is pretty much just an observation, since S is a subgroup its compliment is simply the union of its cosets

$$S^c = \bigcup_{g \in G \setminus S} L_g(S)$$

which is a union of open sets since we have established S is open and L_g is a diffeomorphism. □

7. (a) Let $\phi : G \to H$ be a morphism of connected Lie groups, and assume $d_e\phi : \mathfrak{g} \to \mathfrak{h}$ is bijective (resp. surjective, injective), show that ϕ is a covering (resp. surjective, an immersion).

Proof. I will first show that for ϕ a morphism of lie groups, $d\phi$ has constant rank. To see this let $g \in G$, then

$$L_{\phi(g^{-1})} \circ \phi \circ L_g = \phi$$

So that by the chain rule,

$$(d_{\phi(g^{-1})} L_{\phi(g)})(d_g\phi)(d_e L_g) = d_e\phi$$

Since $L_{\phi(g)}$ and L_g are diffeomorphisms, we have that $\text{rank}(d_{\phi(g^{-1})} L_{\phi(g)})(d_g\phi)(d_e L_g) = \text{rank}\, d_g\phi$, and the above equation shows that $\text{rank}(d_{\phi(g^{-1})} L_{\phi(g)})(d_g\phi)(d_e L_g) = \text{rank}\, d_e\phi$, taken together this gives us

$$\text{rank}\, d_g\phi = \text{rank}\, d_e\phi$$

and since $g \in G$ was arbitrary, ϕ has constant rank as required.

Morphisms of Lie groups having constant rank suffices to show that $d_e\phi$ being bijective (resp. surjective injective) implies that ϕ is a local diffeomorphism (by the inverse function theorem) (resp. submersion, immersion by the submersion and immersion theorems). In the case where $d_e\phi$ is surjective, ϕ is a submersion, so that $\phi(G) \subset H$ is an open subset, and since $\phi(G) \subset H$ is a subgoup, by the same coset argument as in (problem 6), $\phi(G)^c \subset H$ is also open, hence $\phi(G) \subset H$ is closed. Since $\phi(G) \subset H$ is clopen and H is connected $\phi(G) = H$.

It remains to show that $\phi : G \to H$ being a local diffeomorphism implies that it is a covering map. Since $H \cong G/\ker\phi$, it suffices to check that $\ker\phi \subset G$ is a discrete group, and that it acts properly discontinuously, the fact it acts freely and smoothly simply follow from G being a Lie group. To see that $\ker\phi$ is discrete, note that if it had a limit point p, then ϕ would not be injective at p, contradicting the fact it is a local diffeomorphism. Then to see that the action of $\ker\phi$ is properly discontinuous, it suffices by translation to check at the identity. Let $g \in \ker\phi \setminus \{e\}$, then let $K \supset \{e\}$ be compact, we need to check that $\{s \in \ker\phi \mid sK \cap K\} \subset \ker\phi$ is finite. To see this, note that $\{s \in \ker\phi \mid sK \cap K\} \subset KK^{-1}$ by definition and is discrete, so we will be done if we can show that KK^{-1} is compact. Note that since K is compact, $K^{-1} := \text{inv}\, K$ is the continuous image of a compact set, hence also compact, and KK^{-1} is the image of the multiplication map $m : K \times K^{-1} \to G$, since the product of compact sets is compact, and the continuous image of compact sets is compact we conclude that KK^{-1} is compact. □

(b) Give an example where $d_e\phi$ is injective but ϕ is not an embedding.

Proof. consider the double cover $SU(2) \to SO(3)$ the derivative is bijective (in particular injective), but since the map is a double cover is is in particular not injective.

Another example is $\phi : \mathbb{R} \to T^2 = R^2/\mathbb{Z}^2$ via $x \mapsto (x, x\sqrt{2})$, then $d_e\phi = (1, \sqrt{2})$ is injective, but since $\{1, \sqrt{2}\}$ are rationally independent by Kroeneckers lemma, $\overline{\phi(\mathbb{R})} = T^2$, and hence the submanifold topology on $\phi(R)$ is not the topology on $\mathbb{R}$. □

8. Prove that if $H_1, H_2 \subset G$ are Lie subgroups, then so is $H_1 \cap H_2$.

Proof. We know that $H_1 \times H_2 \hookrightarrow G \times G$ is a Lie subgroup, since we can take its inclusion to the product to be the product of the inclusions into G. Since G is a manifold it is Hausdorff, so the diagonal $\Delta := \{(g,g) \mid g \in G\} \subset G \times G$ is closed, moreover it is a subgroup so it is a closed Lie subgroup by the closed subgroup theorem. It follows that $\Delta \cap (H_1 \times H_2)$ is a closed subgroup of $H_1 \times H_2$, but $\Delta \cap (H_1 \times H_2)$ is the image of the diagonal map on $H_1 \cap H_2$, then taking the projection map $\pi_1 : G \times G \to G, (g,h) \mapsto g$, we get

$$H_1 \cap H_2 \cong \Delta \cap (H_1 \times H_2) \xrightarrow{\iota} G \times G \xrightarrow{\pi_1} G$$

and since π_1 is an injective immersion on Δ, its an injective immersion of $\iota(\Delta \cap (H_1 \times H_2))$, hence this realizes $H_1 \cap H_2$ as a Lie subgroup. $\square$

9. Let G be the identity component of the affine group of the real line consisting of transformations $x \mapsto xt + s$, with $t > 0$ and $s \in \mathbb{R}$, identified with matrices of the form $\begin{pmatrix} t & s \\ 0 & 1 \end{pmatrix}$, show that G is not unimodular.

Proof. To show that G is not unimodular, we simply need to provide some $g \in G$ with $\chi(g) \neq 1$. To do so, we first need to compute the lie algebra. As a manifold, $G = \left\{ \begin{pmatrix} t & s \\ 0 & 1 \end{pmatrix} \mid t > 0, s \in \mathbb{R} \right\}$ has coordinates s, t, and whence the smooth one parameter paths through the origin are of the form for $\gamma_j(0) = 1$ smooth paths in $\mathbb{R}$

$$\gamma(t) = \begin{pmatrix} \gamma_1(t) & \gamma_2(t) \\ 0 & 1 \end{pmatrix}$$

Since the paths are smooth, we may write them as $\gamma_1(t) = 1 + at + \mathcal{O}(t^2)$ and $\gamma_2(t) = 1 + bt + \mathcal{O}(t^2)$, it follows that $\frac{d}{dt}|_{t=0}\gamma(t) = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}$, so that $\mathfrak{g} = \operatorname{span}\{e_{11}, e_{12}\}$. So letting $g = \begin{pmatrix} t & s \\ 0 & 1 \end{pmatrix}$, we compute

$$\begin{aligned} \mathrm{Ad}_g(e_{11}) &= e_{11} - se_{12} \\ \mathrm{Ad}_g(e_{12}) &= te_{12} \end{aligned}$$

So that

$$\chi(g) = \left| \det \begin{pmatrix} 1 & 0 \\ -s & t \end{pmatrix} \right| = t$$

So that for $g = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ we have $\chi(g) = 2 \neq 1$. $\square$

10. Prove that if G is a connected Lie group then every ad-invariant inner product on $\mathfrak{g}$ is also Ad-invariant.

Proof. Let $\langle -, - \rangle$ be an ad-invariant inner product, we first show that $\{g \in G \mid \langle -, - \rangle \text{ is } \mathrm{Ad}_g \text{ invariant}\}$ is closed. To prove this, we use that $\langle -, - \rangle$ is bilinear, hence we can take the limit in both arguments, and that $g \mapsto \mathrm{Ad}_g$ is continuous, so that if $\{g_n\} \subset \{g \in G \mid \langle -, - \rangle \text{ is } \mathrm{Ad}_g \text{ invariant}\}$ and $g_n \to g \in G$, then

$$\langle \xi, \eta \rangle = \lim_{n \to \infty} \langle \mathrm{Ad}_{g_n} \xi, \mathrm{Ad}_{g_n} \eta \rangle = \langle \lim_{n \to \infty} \mathrm{Ad}_{g_n} \xi, \lim_{n \to \infty} \mathrm{Ad}_{g_k} \eta \rangle = \langle \mathrm{Ad}_g \xi, \mathrm{Ad}_g \eta \rangle$$

Thus since G is connected, it will suffice to show that $\langle -, - \rangle$ is Ad-invariant on the image of the exponential map, since this is an open neighborhood of the identity in G. Let $\xi \in \mathfrak{g}$, then

$$\langle \mathrm{Ad}_{\exp \xi} \eta, \mathrm{Ad}_{\exp \xi} \zeta \rangle = \langle \exp(\mathrm{ad}_\xi)(\eta), \exp(\mathrm{ad}_\xi)(\zeta) \rangle$$

So it will suffice to show that using ad-invariance we can conclude $\langle \exp(\operatorname{ad}_\xi)(\eta), \exp(\operatorname{ad}_\xi)(\zeta)\rangle = \langle \eta, \zeta\rangle$, ad invariance tells us that $\langle \operatorname{ad}_\xi \eta, \zeta\rangle = -\langle \eta, \operatorname{ad}_\xi \zeta\rangle$, so that $\operatorname{ad}_\xi^* = -\operatorname{ad}_\xi$, so that $\exp(\operatorname{ad}_\xi)^* = \exp(-\operatorname{ad}_\xi)$. Applying this we get

$$\langle \exp(\operatorname{ad}_\xi)(\eta), \exp(\operatorname{ad}_\xi)(\zeta)\rangle = \langle \eta, \exp(-\operatorname{ad}_\xi)\exp(\operatorname{ad}_\xi)(\zeta)\rangle = \langle \eta, \zeta\rangle$$

□

11. $T \subset SU(2)$, the subgroup of diagonal matrices is a maximal torus. So is $T' = SO(2) \subset SU(2)$, find all $a \in SU(2)$ such that $\operatorname{Ad}_a(T) = T'$.

Proof. We can find such $a \in SU(2)$ by diagonalizing rotation matrices, let $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, then $\det(R(\theta) - \lambda) = \lambda^2 - 2\lambda\cos\theta + 1$, which using the quadratic formula has roots $e^{i\theta}, e^{-i\theta}$. We find the eigenvectors corresponding to these roots, we get $e^{i\theta}$ has eigenvector $e_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1 \end{pmatrix}$, and $e^{-i\theta}$ has eigenvector $e_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ -i \end{pmatrix}$ so that

$$\begin{pmatrix} \mathbf{e_1} & \mathbf{e_2} \end{pmatrix} \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \begin{pmatrix} \mathbf{e_1} & \mathbf{e_2} \end{pmatrix}^{-1} = R(\theta)$$

Since these are the eigenvectors for any rotation matrix, the conjugating elements $\operatorname{Ad}_a(T) = T'$ are exactly

$$a \in \left\{ \begin{pmatrix} s\mathbf{e_1} & t\mathbf{e_2} \end{pmatrix}, \begin{pmatrix} s\mathbf{e_2} & t\mathbf{e_1} \end{pmatrix} \mid s, t \in \mathbb{C} \right\} \cap SU(2)$$

We parameterize this set a bit more nicely, by noticing that first off if $a = \begin{pmatrix} \mathbf{f_1} & \mathbf{f_2} \end{pmatrix}$, then since $a \in SU(2)$, we have $||f_j|| = 1$, whence $|s| = |t| = 1$, and that if we scale f_1 by $s \in \mathbb{R}$, then to preserve the determinant, we must multiply f_2 by s^{-1}. Thus we find that using the definitions for e_1, e_2 from above:

$$\{a \in SU(2) \mid \operatorname{Ad}_a(T) \subset T'\} = \left\{ \begin{pmatrix} s\mathbf{e_1} & s^{-1}\mathbf{e_2} \end{pmatrix}, \begin{pmatrix} s\mathbf{e_2} & s^{-1}\mathbf{e_1} \end{pmatrix} \mid s \in \mathbb{C}, |s| = 1 \right\}$$

□

12. Let $H \subset SO(3)$ be the subgroup of diagonal matrices.

(a) Show that $H \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Proof. Since $g \in H$, then since $g^T = g$ is orthogonal, we have $g = g^{-1}$ which implies the diagonal entries must either be ± 1, moreover since $\det g = 1$, there must be an even number of -1 entries. Such matrices can be enumerated

$$1, h_1 = \operatorname{diag}(-1,-1,1), h_2 = \operatorname{diag}(-1,1,-1), h_3 = \operatorname{diag}(1,-1,-1)$$

To see $H \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, we use that H has order 4 and is a subgroup of $SU(2)$, and that the only groups of order 4 are $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$, since every element of H has order 2, we have $H \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. □

(b) Show that H is maximal abelian.

Proof. It will suffice to show that there is no $g \in SO(3)$ such that g commutes with all of H, equivalently, that the only matrices commuting with H are diagonal, to see this let $g \in SO(3)$, then we can write g in block diagonal form as $g = \begin{pmatrix} A & \mathbf{b} \\ \mathbf{a}^T & c \end{pmatrix}$ for A a 2x2 matrix, $c \in \mathbb{R}$, and a, b 2×1 column vectors. For g to commute with H we require

$$(-h_1)g(-h_1) = h_1 g h_1^{-1} = g$$

We use the block form to easily compute this as

$$(-h_1)g(-h_1) = \begin{pmatrix} A & \mathbf{b} \\ -\mathbf{a}^T & -c \end{pmatrix}(-h_1) = \begin{pmatrix} A & -\mathbf{b} \\ -\mathbf{a}^T & c \end{pmatrix}$$

So that the vectors $a = -a = 0$ and $b = -b = 0$, we may similarly write g in block form $g = \begin{pmatrix} c & \mathbf{a^T} \\ \mathbf{b} & A \end{pmatrix}$ for A, b, a, c of the same form. The same calculation conjugating by h_3 gives us again that $a = b = 0$, taken together this tells us that all the off diagonal entries of g must be zero, hence the only matrices commuting with H are diagonal so that H is maximal abelian. $\square$

13. Let $\pi : \mathfrak{sl}(2,\mathbb{C}) \to \mathrm{End}(V)$ be the irreducible $\mathfrak{sl}(2,\mathbb{C})$ representation of dimension $n+1$, denote $\tilde{V} := \mathrm{End}(V)$, then define

$$\begin{aligned} \tilde{\pi} : \mathfrak{sl}(2,\mathbb{C}) &\to \mathrm{End}(\tilde{V}) \\ \xi &\mapsto [\pi(\xi), -] \end{aligned}$$

determine the multiplicities of the irreps $V(k)$ in $\tilde{V}$

Proof. We use the third method provided in the course notes, i.e. determining the eigenspace decomposition of $\tilde{\pi}(h)$ on $\ker \tilde{\pi}(e)$, so that the multiplicity of the eigenvalue k gives $\frac{1}{k+1} \dim U$, where U is the direct sum of all $V(k)$ type irreps, equivalently, the multiplicity of the eigenvalue counts the number of $V(k)$ type irreps.

We first notice that $\pi(e)^j \in \ker \tilde{\pi}(e)$ for all j (this gives us $n+1$ linear independent elements $\{1, \pi(e), \dots, \pi(e)^n\}$), we would like to check that we get equality. To do so it suffices to check that there are at most $n+2$ distinct elements in $\mathrm{End}(V)$ commuting with $\pi(e)$. Suppose that A were such an element, then since $C\pi(e)^j(v_r) = v_{r-j}$ for some constant C, we find that

$$Av_{n-j} = AC\pi(e)^j(v_n) = C\pi(e)^j Av_n$$

so that A is determined by its action on v_n, up to scaling there are only $n+1$ such choices. Now, we can use the relation $[\pi(e), \pi(h)] = -2\pi(e)$ to compute $\tilde{\pi}(h)$, by anticommutativity and recursion we find that

$$\tilde{\pi}(h)(\pi(e)^j) = [\pi(h), \pi(e)^j] = 2j\pi(e)^j$$

Thus we find that the multiplicity of $V(k)$, which is equal to the multiplicity of eigenvalue k is:

$$\text{multiplicity of } V(k) = \begin{cases} 1 & 0 \le k \le 2n \text{ is even} \\ 0 & \text{else} \end{cases}$$

$\square$